

T5Gemma v1 vs Gemma-2-2B-IT vs Qwen3-4B on TReB

Three-way comparison — Claude Sonnet judge

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TL;DR

Pretraining quality swamps architecture at the 2–5B scale. We ran two Gemma-2-lineage siblings — encoder-decoder T5Gemma v1 vs pure decoder-only Gemma-2-2B-IT (same base weights, same SFT/RLHF recipe, only architecture differs) — and a contemporaneous decoder-only from a different lineage (Qwen3-4B-Instruct-2507) on 248 stratified TReB English/TCoT samples.

Variant	Params	Arch	Released	Mean	CORRECT%	Sole wins
T5Gemma v1 2B-2B UL2-IT	5.6 B	enc-dec	Jun 2025	4.10	32.3%	15
Gemma-2-2B-IT	2.6 B	dec-only	Jul 2024	3.81	30.2%	13
Qwen3-4B-Instruct-2507	4.0 B	dec-only	Aug 2025	4.99	33.1%	27

- Within the Gemma-2 lineage, encoder-decoder roughly ties decoder-only.** T5Gemma edges Gemma-2-2B-IT by 0.29 on mean score and 2.1 pp on CORRECT — small. The clean architectural bump shows up only on the retrieval family (Table_Query, Table_Retrieval, Table_Domain-Ops). Elsewhere enc-dec is a wash.
- Qwen3-4B beats both**, despite being 30% smaller than T5Gemma and from an entirely different pretraining culture. On 55 samples where exactly one model got CORRECT, Qwen3 claims 27 (49%), T5Gemma 15 (27%), Gemma-2 13 (24%).
- Qwen3’s 45.6% TRUNCATED rate** is a chain-of-thought-length artifact, not reasoning failure. Against our 256-token cap, Qwen3’s verbose reasoning hits the ceiling almost half the time — yet it still wins overall. Raising to 1024 tokens would likely widen the lead further.
- 47.6% of samples are all-3-wrong** — the 2–5B-class ceiling on TReB at TCoT difficulty; architecture choice doesn’t move it.

1 Release timeline & lineage

Jul 2024	→	Jun 2025	Aug 2025
Gemma-2-2B-IT 2.6 B dec-only (Google)	+encoder +UL2 adapt +retrain	T5Gemma v1 5.6 B enc-dec (Google, same SFT)	Qwen3-4B-Inst-2507 4.0 B dec-only (Alibaba)
same Gemma-2 pretraining base		different	pretraining lineage

T5Gemma v1 is not a from-scratch encoder-decoder. Per the v1 tech report (arXiv:2504.06225), Google took the Gemma-2-2B pretrained decoder, duplicated & re-initialized it as an encoder, and adapted the pair with UL2 denoising objectives, then ran their usual SFT + RLHF + model-merging IT pipeline. So the T5Gemma v1 vs Gemma-2-2B-IT pair is *the same pretraining + the same post-training*, separated only by “bolt on an encoder and do the UL2 adapt.” Qwen3-4B is a separate lineage

entirely — apples-vs-oranges on pretraining, but it’s the contemporaneous dec-only of comparable size so it’s the obvious sanity check.

2 Setup

- **Samples:** 248 stratified English/TCoT samples from TReB, pre-filtered to prompts ≤ 4000 T5Gemma tokens (SWA-bug gate). All three variants evaluated on the same 248 sample ids.
- **Judge:** Claude Sonnet 4.6 via parallel Claude Code subagents, 12-mode failure taxonomy (CORRECT, PARTIAL, FORMAT, HALLUCINATION, WRONG_SELECTION, NUMERIC, REASONING, REFUSAL, EMPTY, MISUNDERSTAND, TRUNCATED, OTHER).
- **Generation:** bf16, greedy, `max_new_tokens=256` for all three.
- **Backend:** T5Gemma via HF `AutoModelForSeq2SeqLM`, Gemma-2 and Qwen3 via HF `AutoModelForCausalLM`. No vLLM (contested-GPU issues).

3 Per-TReB-category breakdown

Counts per TReB capability category with 3-way correctness patterns (T=T5Gemma CORRECT, G=Gemma-2 CORRECT, Q=Qwen3 CORRECT, dash = not-CORRECT):

Category	n	all3	none	T-only	G-only	Q-only	TG	TQ	GQ
NLU	98	19	39	5	7	10	6	6	6
Table Understanding	73	5	35	6	5	3	12	4	3
Table Basic Ops	35	3	18	1	0	9	0	3	1
Table Computational Ops	23	5	11	2	0	4	1	0	0
Data Analysis	18	0	15	1	1	1	0	0	0
Advanced Data Analysis	1	0	0	0	0	0	—	—	—

Table Understanding (73 samples) is where the Gemma-2 lineage earns its keep. T5G sole wins 6 + T5G+G2-but-not-Qwen wins 12 = **18 of 73 samples** where the Gemma family beats Qwen3. These are tasks where Qwen3’s verbose reasoning gets truncated mid-thought.

Table Basic Ops (35 samples) is Qwen3’s domain: 9 sole wins to 1 (T5G) and 0 (G2). When Qwen3 finishes, it does this family better.

NLU (98 samples) is mixed — Qwen3 edges ahead on sole wins (10 vs 5/7) but totals are close.

Data Analysis (18 samples) is 15/18 all-wrong — nobody at this scale can compute a correlation coefficient or run a distribution test reliably.

Per-category dominant failure modes

Category (n)	T5Gemma v1	Gemma-2-2B-IT
NLU (98)	CORRECT=36, WRONG_S=27, TRUNC=18	CORRECT=38, WRONG_S=25, TRUNC=11
Table Understanding (73)	CORRECT=27, WRONG_S=20, PARTIAL=18	CORRECT=25, PARTIAL=16, WRONG_S=15
Table Basic Ops (35)	WRONG_S=10, CORRECT=7, TRUNC=7	WRONG_S=13, NUMERIC=6, PARTIAL=4
Table Computational Ops (23)	CORRECT=8, TRUNC=7, NUMERIC=5	CORRECT=6, TRUNC=5, NUMERIC=5
Data Analysis (18)	TRUNC=8, WRONG_S=3, REASON=2	TRUNC=9, REFUSAL=3, EMPTY=3

Qwen3’s modal failure in every category is TRUNCATED. T5Gemma’s is WRONG_SELECTION. Gemma-2’s is WRONG_SELECTION with a distinctive REFUSAL + EMPTY tail (says “I can’t access that table” more than either sibling).

4 What this says about encoder-decoder architecture

Against the *cleanest possible architectural comparator* (Gemma-2-2B-IT — same pretraining base, same IT recipe), T5Gemma v1’s encoder-decoder structure:

- **Wins the retrieval family** (Table_Retrieval +1.7, Table_Query +1.5, Table_Domain-Ops +1.7 on mean score). Mechanistically consistent with cross-attention: the decoder can re-attend to the encoded table at every generation step.
- **Ties or loses everywhere else.** Mean score +0.29, CORRECT% +2.1 pp — nothing to write home about when you’re paying 2× the parameters for it.

Against the *newer, different-lineage comparator* (Qwen3-4B-Instruct-2507), the architectural edge is invisible. Qwen3 beats both Gemma variants on mean score and CORRECT%, and does so with 30% fewer params than T5Gemma. That tells us pretraining + post-training quality is more important than architecture at this parameter scale.

5 Recommended follow-ups

1. **Rerun Qwen3-4B at max_new_tokens=1024** — would likely convert a big chunk of its 113 TRUNCATED outcomes into CORRECT. Cheap (~\$0.50).
2. **Qwen3-14B** for same-lineage scaling comparison, to see if the 47.6% all-wrong bucket is a 2–5B scale ceiling or a benchmark-difficulty floor.
3. **Gemma-2-9B-IT** as a param-equal Gemma-2 decoder-only (closer to T5Gemma’s 5.6B total params). Would sharpen the enc-dec-vs-dec-only comparison within Gemma lineage.
4. **Relax the 4K-token filter** if T5Gemma v1’s SWA bug doesn’t trigger above 4094 — would recover the Multi-step_* family that’s currently reduced to $n=1$.

Cost

~\$4 of RunPod time across three pod runs + two aborted vLLM attempts. Sonnet judge pass: free (parallel Claude Code subagents). ~12 min of wall-clock judge time for 744 judgments.

Appendix — 5 illustrative examples

(1) T5Gemma’s encoder-decoder wins a retrieval task

Task: Table_Title_Naming id: feeb1c38...

field	content
Q	“Please provide a title for the table based on its content.” (table: 2014 ICC World Cricket League Division Four stats)
Gold	The title is 2014 ICC World Cricket League Division Four
T5Gemma	CORRECT (10) — {"answer": "2014 ICC World Cricket League Division Four"} — exact match
Gemma-2	PARTIAL (8) — {"answer": "...Division Four Information"} — appended “Information”
Qwen3	PARTIAL (8) — {"answer": "...Division Four: Event Details"} — appended “: Event Details”

All three see the table content, but T5Gemma commits to the exact gold phrasing. Gemma-2 and Qwen3 both add descriptive suffixes — a subtle behavior that costs 2 points.

(2) Qwen3 wins a task needing semantic vocabulary

Task: Table_Query id: 0e92a63b...

field	content
Q	which column is named "OT" in the table? — gold expects <i>column index</i>
Gold	This column is the 2th one. (number_answer: 2)
T5Gemma	PARTIAL — {"answer": "OT"} — answered the column <i>name</i> not the index
Gemma-2	WRONG SELECTION — same trap, answered "OT"
Qwen3	CORRECT — {"answer": "second"} — correctly reads “which column” as positional

Both Gemmas parse the question the same (wrong) way; Qwen3’s instruction-following catches that “which column is named X” means “give me the ordinal.”

(3) Qwen3 truncation casualty — had the answer, ran out of tokens

Task: Table_Fact_Checking id: 1fb344ea...

field	content
Q	“the model with 90% after 1 year have 80% after 3 year” — fact-check Entailment vs Contradiction
Gold	Entailment
T5Gemma	CORRECT — identifies hr-3utgb meets criteria, outputs Entailment
Gemma-2	CORRECT — same logic, Entailment
Qwen3	TRUNCATED — <i>reasoning</i> identifies hr-3utgb at 90%/80%, supports Entailment, but generation stops before emitting {"answer": ...}

Qwen3 had the right answer but burned 256 tokens on reasoning. This is the 113/248-sample pattern that a larger `max_new_tokens` would fix.

(4) All 3 fail at the 2–5B ceiling

Task: Table_Correlation_Analysis id: 5ec3400a...

field	content
Q	“What is the correlation between the ‘number of branches’ and ‘total assets’ for a bank? Provide the correlation coefficient as evidence.”
Gold	Strong positive correlation, 0.97
T5Gemma	FORMAT — explains Pearson formula methodology, never computes the number
Gemma-2	TRUNCATED — cuts off mid-formula
Qwen3	TRUNCATED — lists variable names, starts Pearson formula, cuts off before computing

All three know the formula. None can actually execute the arithmetic over tabular data and emit the coefficient. Representative of the 47.6%-all-wrong bucket: the Data Analysis family is above what this parameter class can do.

(5) Gemma-2-2B-IT wins a classification task

Task: Robustness_Evaluation **id:** 8f7d6ce7...

field	content
Q	NLI-style: (A) Dana Reeve widow of Christopher Reeve... vs (B) Dana Reeve... → entailment or not_entailment
Gold	not_entailment
T5Gemma	WRONG_SELECTION — returns entailment (wrong)
Gemma-2	CORRECT — not_entailment
Qwen3	TRUNCATED — gets to “Step 4: Check if (A) entails (B):” and stops

Gemma-2 is particularly strong on Robustness_Evaluation (NLI labels) — commits to a label efficiently. T5Gemma hedges toward entailment; Qwen3 over-reasons itself out of an answer. Gemma-2 wins 5/17 of these samples uniquely.